

Chapter 1

INTRODUCTION

Smart agriculture is an advanced farming approach that uses IoT technology to improve efficiency and resource management. This project focuses on monitoring soil conditions using sensors and controlling irrigation through the Blynk mobile application. The system uses an ESP8266 microcontroller to collect and transmit real-time data such as soil moisture, temperature, and humidity. Based on the soil condition, the water pump can be controlled automatically or manually through the mobile app. This helps in reducing manual effort and supports efficient farming practices.

1.1 Brief history of project

Earlier, agriculture mainly depended on traditional irrigation methods where farmers manually monitored soil and supplied water to crops. These methods often caused water wastage and required more labor. With the development of sensors, microcontrollers, and IoT technology, smart agriculture systems became possible. Modern irrigation systems now use real-time monitoring and automation to improve efficiency and reduce manual effort. This project is developed based on these advancements to provide a smart and automated soil monitoring and irrigation control system.

1.2 Modern agricultural system

Modern agriculture systems use advanced technologies such as IoT, sensors, automation, and smart monitoring to improve farming efficiency and productivity. These systems help farmers monitor soil conditions, temperature, humidity, and irrigation in real time. Automated irrigation systems reduce water wastage and minimize manual effort. Mobile applications and cloud platforms allow remote monitoring and control of agricultural activities. Modern agriculture promotes efficient resource management, higher crop yield, and sustainable farming practices.

Chapter 2

Problem Statement

2.1 Description

Traditional irrigation systems mainly depend on manual monitoring and operation, which can lead to inefficient irrigation and wastage of resources. Farmers may not always know the exact soil condition, temperature, or humidity level required for crops. Lack of real-time monitoring makes irrigation management difficult and increases manual effort. Existing systems also provide limited remote access and automation facilities. Therefore, there is a need for an IoT-based smart agriculture system that can monitor soil conditions in real time and control irrigation efficiently through automation and mobile applications.

2.2 Challenge Statement

Developing a smart agriculture system involves several challenges such as obtaining accurate sensor readings under different environmental conditions, maintaining stable IoT connectivity, and ensuring reliable automation of irrigation systems. Power management and continuous monitoring in outdoor agricultural environments are also difficult. Integrating multiple sensors and controlling the system efficiently through a mobile application while keeping the system cost-effective is another major challenge. The project aims to address these issues by creating a reliable and user-friendly smart monitoring system

Chapter 3

3.1 Design Thinking Process

a) Empathize:

Discussions with farmers and users revealed problems such as inefficient irrigation, lack of real-time monitoring, and difficulty in managing irrigation manually.

b) Define:

The key needs identified were automatic irrigation control, real-time soil monitoring, remote access through mobile applications, and efficient resource management.

c) Ideate:

Multiple ideas were generated using IoT sensors, mobile applications, and automation techniques. The final idea selected was an IoT-based smart soil monitoring and irrigation control system using Blynk.

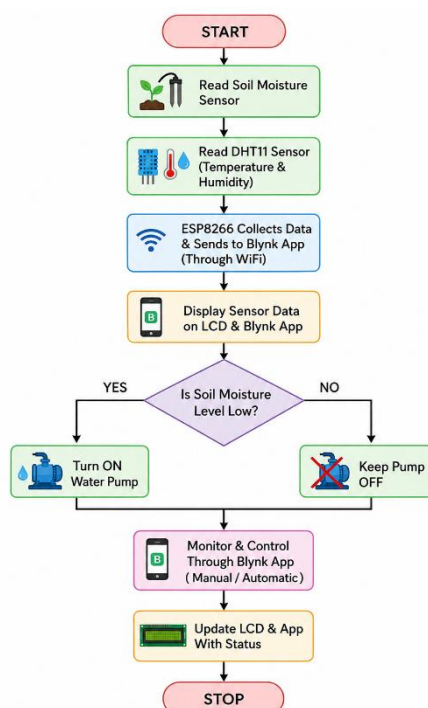
d) Prototype:

A prototype was developed using ESP8266, soil moisture sensor, DHT11 sensor, relay module, LCD display, and water pump integrated with the Blynk application.

e) Test:

The system was tested for monitoring soil conditions and controlling the pump both automatically and manually. The prototype successfully displayed sensor data and controlled irrigation efficiently.

3.2 Methodology

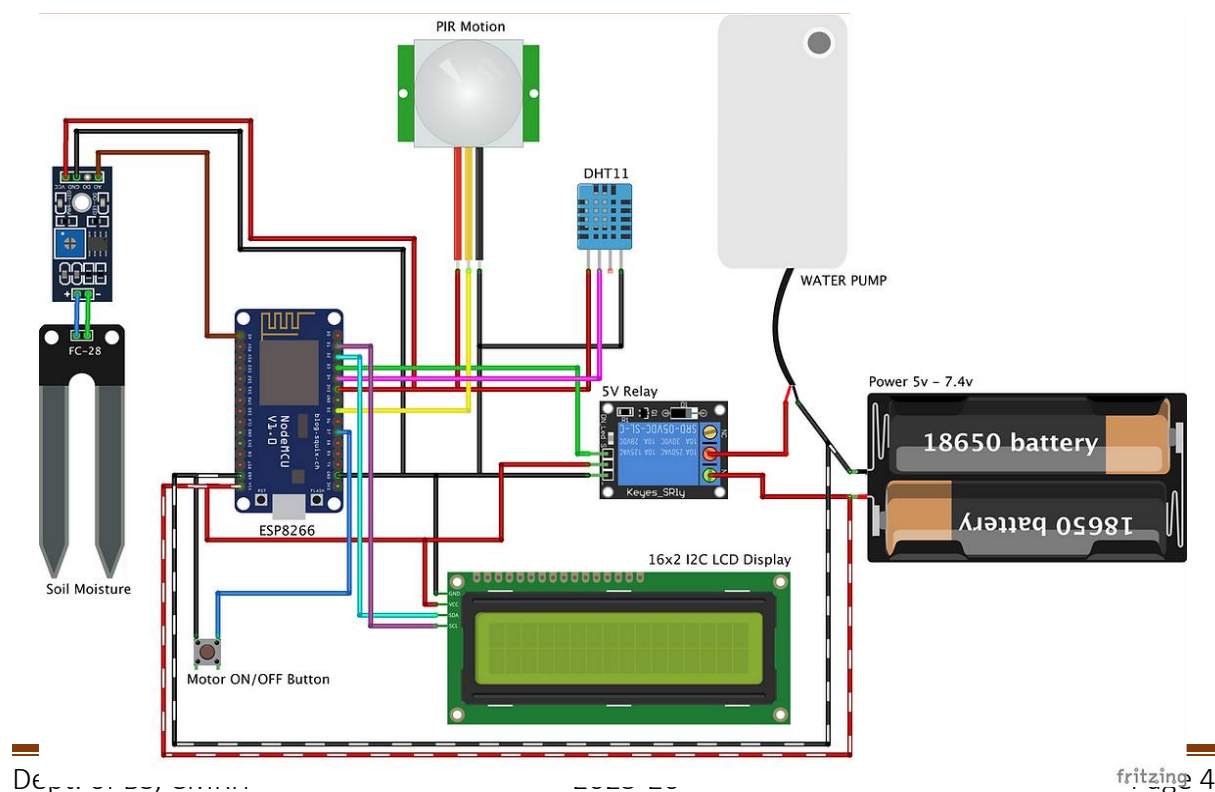


3.3 Prototype Description

3.3.1 Materials Used

- ESP8266 (NodeMCU)
- Soil Moisture Sensor
- DHT11 Sensor
- PIR Motion Sensor
- Relay 5V Module
- 16×2 I2C LCD Display
- Tactile Button
- Small Water Pump
- Breadboard
- Connecting Wires
- 18650 Battery

3.3.2 Circuit Diagram



Chapter 4

Implementation

```
//Include the library files

#include <LiquidCrystal_I2C.h>
#define BLYNK_PRINT Serial
#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>
#include <DHT.h>

//Initialize the LCD display
LiquidCrystal_I2C lcd(0x3F, 16, 2);

char auth[] = "    "; //Enter your Blynk Auth token
char ssid[] = "    "; //Enter your WIFI SSID
char pass[] = "    "; //Enter your WIFI Password

DHT dht(D4, DHT11); //(DHT sensor pin,sensor type)  D4 DHT11 Temperature Sensor
BlynkTimer timer;

//Define component pins
#define soil A0      //A0 Soil Moisture Sensor
#define PIR D5       //D5 PIR Motion Sensor
int PIR_ToggleValue;

void checkPhysicalButton();
int relay1State = LOW;
int pushButton1State = HIGH;
#define RELAY_PIN_1    D3    //D3 Relay
#define PUSH_BUTTON_1  D7    //D7 Button
#define VPIN_BUTTON_1  V12

//Create three variables for pressure
double T, P;
char status;
```

```
void setup() {
  Serial.begin(9600);
  lcd.begin();
  lcd.backlight();
  pinMode(PIR, INPUT);

  pinMode(RELAY_PIN_1, OUTPUT);
  digitalWrite(RELAY_PIN_1, LOW);
  pinMode(PUSH_BUTTON_1, INPUT_PULLUP);
  digitalWrite(RELAY_PIN_1, relay1State);

  Blynk.begin(auth, ssid, pass, "blynk.cloud", 80);
  dht.begin();

  lcd.setCursor(0, 0);
  lcd.print("  Initializing  ");
  for (int a = 5; a <= 10; a++) {
    lcd.setCursor(a, 1);
    lcd.print(".");
    delay(500);
  }
  lcd.clear();
  lcd.setCursor(11, 1);
  lcd.print("W:OFF");
  //Call the function
  timer.setInterval(100L, soilMoistureSensor);
  timer.setInterval(100L, DHT11sensor);
  timer.setInterval(500L, checkPhysicalButton);
}

//Get the DHT11 sensor values
void DHT11sensor() {
  float h = dht.readHumidity();
  float t = dht.readTemperature();

  if (isnan(h) || isnan(t)) {
    Serial.println("Failed to read from DHT sensor!");
    return;
  }
  Blynk.virtualWrite(V0, t);
  Blynk.virtualWrite(V1, h);

  lcd.setCursor(0, 0);
  lcd.print("T:");
  lcd.print(t);
```

```
lcd.setCursor(8, 0);  
lcd.print("H:");  
lcd.print(h);  
  
}
```

Chapter 5

Results and Analysis

User Testing & Feedback

Participants:

Testing was conducted with classmates, friends, and faculty members to evaluate the system performance and usability.

Quantitative Results:

- Soil moisture data was updated in real time on the Blynk application.
- The water pump responded successfully in both automatic and manual modes.
- Sensor readings were displayed accurately on the LCD display.
- The system showed stable IoT connectivity during testing.

Qualitative Feedback:

- Users found the mobile control feature easy and convenient to use.
- Faculty appreciated the automation and real-time monitoring features.
- The system was considered useful for smart agriculture and efficient irrigation management.

Chapter 6

Conclusion & Future Work

The IoT-Based Smart Soil Monitoring and Irrigation Control System provides an efficient solution for modern agriculture by enabling real-time monitoring and automated irrigation control. The system reduces manual effort, supports efficient resource usage, and allows users to monitor and control irrigation remotely through the Blynk application. The integration of sensors and IoT technology improves farming efficiency and promotes smart agriculture practices.

Future Work:

- AI-based crop and soil analysis.
- Solar-powered irrigation system.
- Advanced mobile notifications and alerts.
- Cloud data storage and analytics.
- Integration with weather forecasting systems.

References

Research Papers

1. S. R. Nandurkar, V. R. Thool, and R. C. Thool, "Design and Development of Precision Agriculture System Using Wireless Sensor Network," IEEE International Conference, 2014.
2. "IoT Based Smart Irrigation Management System for Environmental Sustainability," ScienceDirect, 202

Online articles

- <https://www.electronicshub.org/iot-based-smart-irrigation-system/>
- https://en.wikipedia.org/wiki/Internet_of_things
- <https://www.arduino.cc/>
- <https://www.blynk.io/>

Annexures

Annexure A – User Feedback Forms

Annexure B – Iteration Notes

Annexure C – Team Roles